

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer, Secret Clearance (Active)

Dallas, TX

June 2025 – Present

- Developed signal processing modules in Python and C++ on embedded systems for defense applications, collaborating with hardware teams to meet strict latency and resource constraints in a classified environment.
- Built and maintained containerized microservices for high-frequency data ingestion and processing using Podman and serverless architecture, ensuring fault-tolerant operation over extended periods with minimal intervention.
- Automated CI/CD pipelines with static code analysis and integration testing across multiple deployment environments, reducing downtime and enabling reliable software delivery to production hardware systems.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

- Designed and deployed highly available Python and TypeScript backend services on AWS Fargate and SageMaker, orchestrating real-time data ingestion from WebRTC video pipelines and LLM-based inference (PyTorch, Llama) across distributed systems running with minimal operator intervention.
- Built CI/CD automation and infrastructure-as-code tooling that enabled reliable, frequent software deployment to production systems with minimal downtime, supporting multiple weekly releases across cloud and containerized environments.
- Developed test execution infrastructure covering unit, integration, and end-to-end scenarios across virtualized and live service environments, significantly improving release confidence and reducing production defect rates.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.